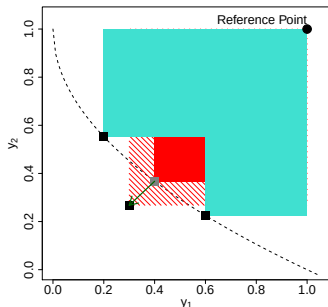
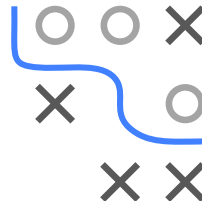


Optimization in Machine Learning

Bayesian Optimization

Multicriteria Bayesian Optimization



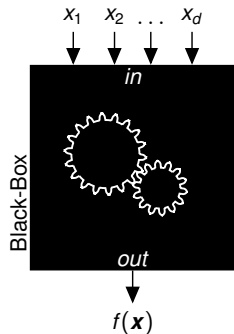
Learning goals

- Multicriteria Optimization
- Taxonomy
- ParEGO, SMS-EGO, EHI

MULTICRITERIA BAYESIAN OPTIMIZATION

$$f : \mathcal{S} \rightarrow \mathbb{R}^m$$

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))$$



- A configuration $\mathbf{x}$ **dominates** ($\prec$) $\tilde{\mathbf{x}}$ if

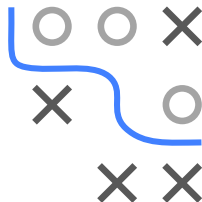
$$\forall i \in \{1, \dots, m\} : f_i(\mathbf{x}) \leq f_i(\tilde{\mathbf{x}})$$

$$\text{and } \exists j \in \{1, \dots, m\} : f_j(\mathbf{x}) < f_j(\tilde{\mathbf{x}})$$

- Set of non-dominated solutions:

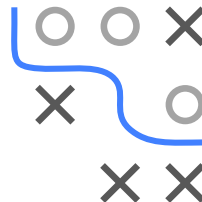
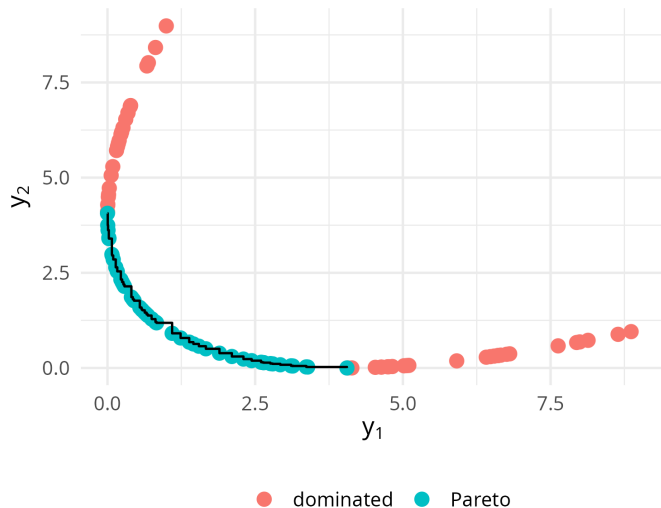
$$\mathcal{P} := \{\mathbf{x} \in \mathcal{S} \mid \nexists \tilde{\mathbf{x}} \in \mathcal{S} : \tilde{\mathbf{x}} \prec \mathbf{x}\}$$

- Pareto set $\mathcal{P}$, Pareto front $\mathcal{F} = f(\mathcal{P})$
- Goal: Find $\hat{\mathcal{P}}$ of non-dominated points that estimates the true Pareto set $\mathcal{P}$



MULTICRITERIA BAYESIAN OPTIMIZATION

Example Pareto front:



MULTICRITERIA BAYESIAN OPTIMIZATION

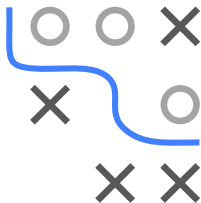
The most popular quality indicator is the hypervolume indicator (also called dominated hypervolume or $\mathcal{S}$ -metric).

The hypervolume, HV, of an approximation of the Pareto front $\hat{\mathcal{F}} = f(\hat{\mathcal{P}})$ can be defined as the combined volume of the dominated hypercubes domHC_r of all solution points $\mathbf{x} \in \hat{\mathcal{P}}$ regarding a reference point $\mathbf{r}$, i.e.,

$$\text{HV}_r(\hat{\mathcal{P}}) := \mu \left(\bigcup_{\mathbf{x} \in \hat{\mathcal{P}}} \text{domHC}_r(\mathbf{x}) \right)$$

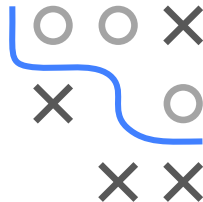
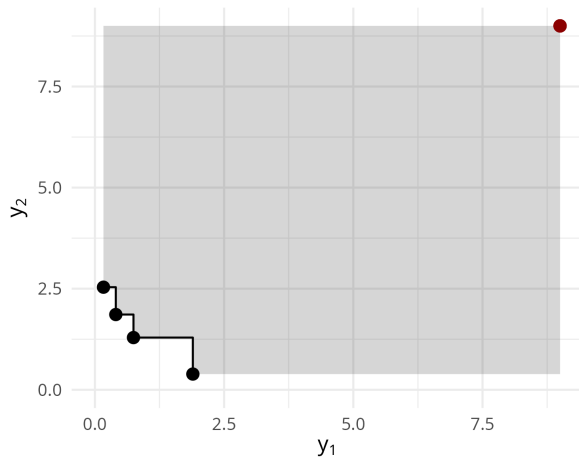
where μ is the Lebesgue measure and the dominated hypercube is given as:

$$\text{domHC}_r(\mathbf{x}) := \{ \mathbf{u} \in \mathbb{R}^m \mid f_i(\mathbf{x}) \leq u_i \leq r_i \forall i \in \{1, \dots, m\} \}$$



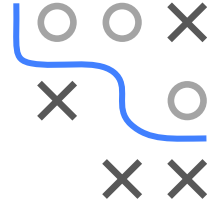
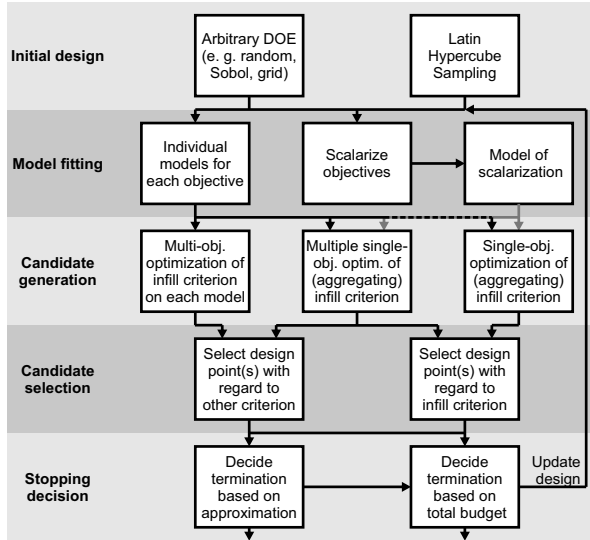
MULTICRITERIA BAYESIAN OPTIMIZATION

Hypervolume example:

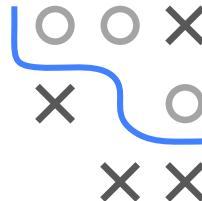
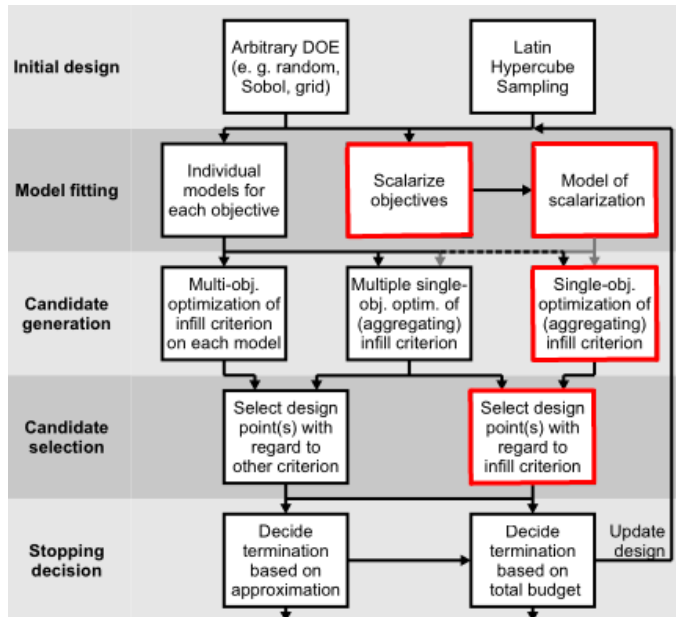


Reference point $\mathbf{r}$ in red, estimated Pareto front $\hat{\mathcal{F}}$ in black,
corresponding $HV_{\mathbf{r}}(\hat{\mathcal{P}})$ is given by the grey area

TAXONOMY



PAREGO



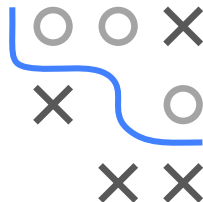
PAREGO

- 1 Scalarize standardized objectives using the augmented Tchebycheff norm

$$\max_{i \in \{1, \dots, m\}} w_i f_i(\mathbf{x}) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x})$$

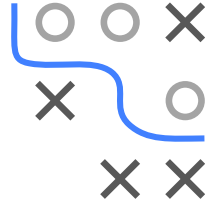
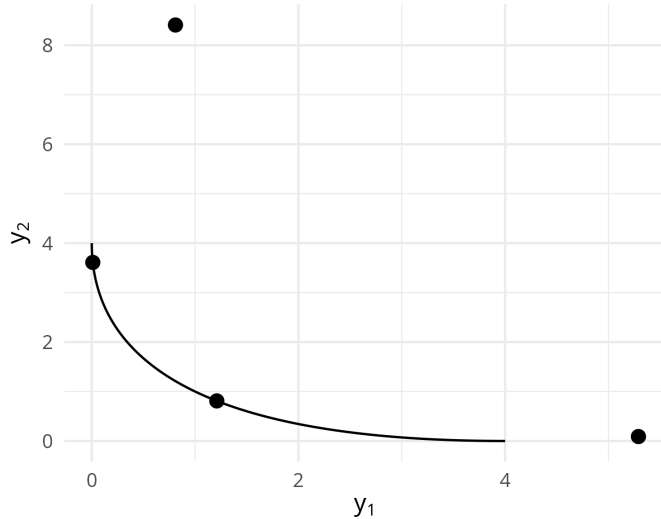
with weight vector $\mathbf{w}$ drawn uniformly from the set of evenly distributed weight vectors $\mathcal{W}$

- 2 Fit SM on the scalarized objective function
- 3 Proceed to use any standard single-objective acquisition function (EI, PI, LCB, ...)



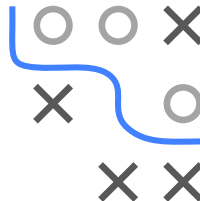
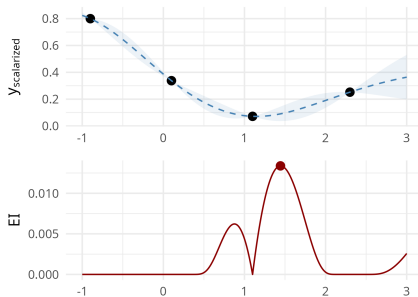
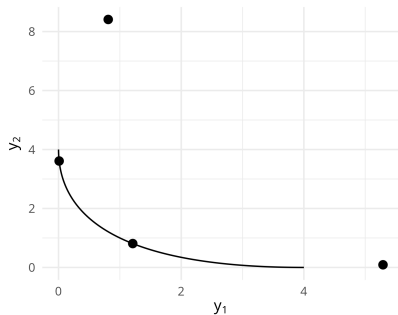
PAREGO

ParEGO Example, initial design and true Pareto front in black ...



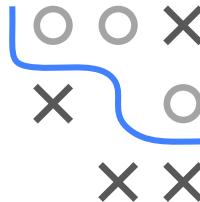
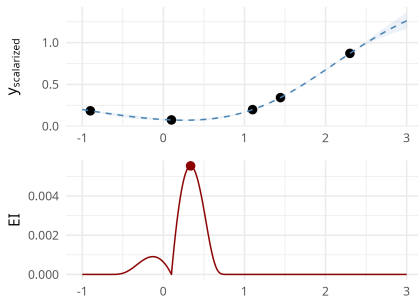
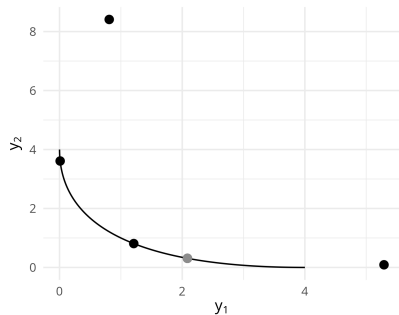
PAREGO

... standardize objectives, obtain scalarized objective via augmented Tchebycheff norm, fit SM and optimize EI ...



PAREGO

... note that the specific scalarization is different at each iteration!



The grey point visualizes the candidate we choose to evaluate in the previous iteration

```
graph TD
    subgraph Initial_design [Initial design]
        A[Arbitrary DOE  
(e. g. random,  
Sobol, grid)]
        B[Latin Hypercube Sampling]
    end

    subgraph Model_fitting [Model fitting]
        C[Individual models for each objective]
        D[Scalarize objectives]
        E[Model of scalarization]
    end

    subgraph Candidate_generation [Candidate generation]
        F[Multi-obj. optimization of infill criterion on each model]
        G[Multiple single-obj. optim. of (aggregating) infill criterion]
        H[Single-obj. optimization of (aggregating) infill criterion]
    end

    subgraph Candidate_selection [Candidate selection]
        I[Select design point(s) with regard to other criterion]
        J[Select design point(s) with regard to infill criterion]
    end

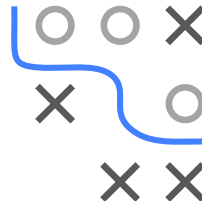
    subgraph Stopping_decision [Stopping decision]
        K[Decide termination based on approximation]
        L[Decide termination based on total budget]
    end

    A --> C
    A --> D
    B --> E
    C --> F
    D --> G
    D --> H
    E --> H
    F --> I
    G --> I
    G --> J
    H --> J
    I --> K
    J --> L
    K --> L
    L --> Update[Update design]
    Update --> B
```

The flowchart illustrates the multi-objective optimization process, organized into five main stages:

- Initial design:** Starts with two methods: "Arbitrary DOE (e. g. random, Sobol, grid)" and "Latin Hypercube Sampling".
- Model fitting:** The process branches into three parallel paths:
 - Path 1: "Individual models for each objective" (highlighted with a red border).
 - Path 2: "Scalarize objectives".
 - Path 3: "Model of scalarization".
- Candidate generation:** Each path from model fitting leads to a corresponding optimization step:
 - From "Individual models": "Multi-obj. optimization of infill criterion on each model".
 - From "Scalarize objectives": "Multiple single-obj. optim. of (aggregating) infill criterion".
 - From "Model of scalarization": "Single-obj. optimization of (aggregating) infill criterion" (highlighted with a red border).
- Candidate selection:** The optimization results lead to selection steps:
 - From "Multi-obj. optimization": "Select design point(s) with regard to other criterion".
 - From "Multiple single-obj. optim.": Both "Select design point(s) with regard to other criterion" and "Select design point(s) with regard to infill criterion" (highlighted with a red border).
 - From "Single-obj. optimization": "Select design point(s) with regard to infill criterion".
- Stopping decision:** The selection process leads to termination decisions:
 - From "Select design point(s) with regard to other criterion": "Decide termination based on approximation".
 - From "Select design point(s) with regard to infill criterion": "Decide termination based on total budget".

The process concludes with a feedback loop: "Update design" leads back to the "Initial design" stage.

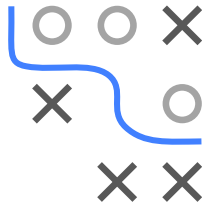
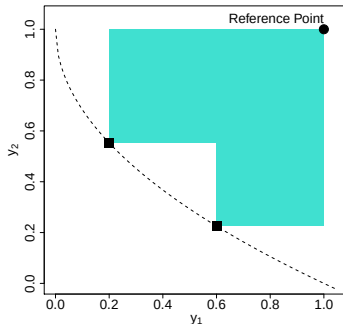


SMS-EGO

Individual models for each objective f_i

Single-objective optimization of aggregating acquisition function:
Calculate contribution of the confidence bound of candidate to the current front approximation

- Calculate LCB for each objective
- Measure contribution with regard to the hypervolume improvement
- For ε -dominated ($\prec_\varepsilon$) solutions, a penalty is added

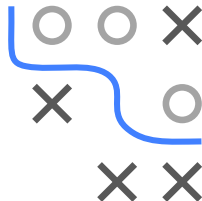
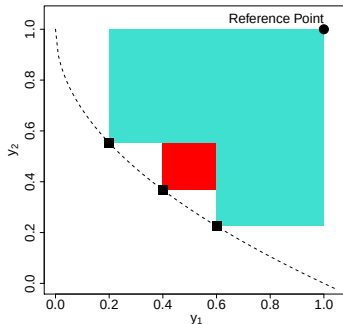


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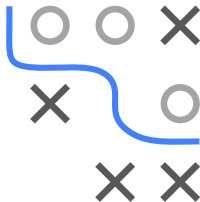
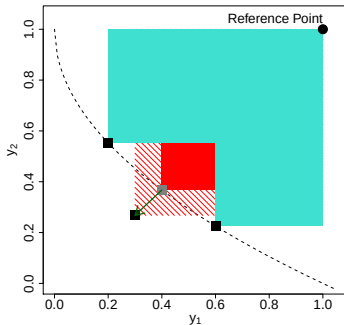


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OUTLOOK

- Many more options exist:
 - Expected Hypervolume Improvement
 - Multi-EGO
 - Entropy based: PESMO, MESMO
 - ...

